CHAPTER 8

Electromagnetic Waves

Sr. No.	Concept	Formulas	Other information
1.	**Displacement current**	Displacement current, $I_D = \varepsilon_0 \frac{d\phi_E}{dt}$ Also $I_D = \varepsilon_0 \frac{d}{dt}(EA) = \varepsilon_0 A \frac{dE}{dt}$ $= \varepsilon_0 A \frac{d}{dt}\left(\frac{V}{d}\right) = \frac{\varepsilon_0 A}{d}\frac{dV}{dt} = C\frac{dV}{dt}$	ε_0 = permittivity of free space. ϕ_E = electric flux. A = area of the capacitor plate. V = electric potential. E = electric field. C = capacitance of a parallel plate capacitor.
2.	**Ampere's circuital law**	$\oint_C \vec{B}.d\vec{l} = \mu_0 I$	B = magnetic field. I = current flowing through the loop.
3.	**Ampere - Maxwell law, (Modified Ampere's circuital law)**	$\oint \vec{B}.d\vec{l} = \mu_0 (I_C + I_D)$ $= \mu_0 \left(I_C + \varepsilon_0 \frac{d\phi_E}{dt}\right)$	I_C = conduction current. I_D = displacement current. μ_0 = permeability of free space.
4.	**Maxwell's equations**	(i) Gauss's law of electrostatics: $\oint_s \vec{E}\cdot d\vec{S} = \frac{q}{\varepsilon_0}$ (ii) Gauss's law of magnetism: $\oint_s \vec{B}\cdot d\vec{S} = 0$ (iii) Faraday's law of electromagnetic induction: $\oint \vec{E}\cdot d\vec{l} = -\frac{d\phi_B}{dt} = -\frac{d}{dt}\left[\oint \vec{B}\cdot d\vec{S}\right]$ (iv) Ampere Maxwell circuital law: $\oint \vec{B}\cdot d\vec{l} = \mu_0\left[I_C + \varepsilon_0 \frac{d\phi_E}{dt}\right]$	E = electric field dS = small area of gaussian surface B = magnetic field q = total charge enclosed by gaussian surface $d\phi_B$ = change in magnetic flux dt = small time interval

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5.	**Mathematical representation of electromagnetic waves**	$\vec{E} = E_y \hat{j} = E_0 \sin(kx - \omega t)\hat{j}$ $= E_0 \sin\left[2\pi\left(\frac{x}{\lambda} - \upsilon t\right)\right]\hat{j}$ $= E_0 \sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right]\hat{j}$ $E_x = E_z = 0$ $\vec{B} = B_z \hat{k} = B_0 \sin(kx - \omega t)\hat{k}$ $= B_0 \sin\left[2\pi\left(\frac{x}{\lambda} - \upsilon t\right)\right]\hat{k}$ $= B_0 \sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right]\hat{k}$ $B_x = B_y = 0$ Propagation constant, $k = \frac{2\pi}{\lambda} = \frac{\omega}{v}$	E_0 = amplitude of electric field. B_0 = amplitude of magnetic field k = propagation constant v = velocity of the wave λ = wavelength of EM wave. t = time. ω = angular frequency υ = frequency of the wave
6.	**Basic properties of electromagnetic waves**	All electromagnetic waves travel in free space with same speed c given by $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \simeq 3 \times 10^8 \text{ ms}^{-1}$ In a material medium of refractive index n, the speed of an electromagnetic wave is given by $v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{c}{n}$ The ratio of the amplitudes of electric and magnetic fields is: $\frac{E_0}{B_0} = c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$	c = speed of light. ε_r = relative permittivity. μ_r = relative permeability. E_0 = amplitude of electric field. B_0 = amplitude of magnetic field $\mu_r = \frac{\mu}{\mu_0}$, $\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$ ε = permittivity of medium μ = permeability of medium
7.	**The average energy density of an electromagnetic wave**	Average energy density of E-field, $u_E = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{2}\varepsilon_0 E_{rms}^2$ Average energy density of B-field, $u_B = \frac{1}{4\mu_0} B_0^2 = \frac{1}{2\mu_0} B_{rms}^2$ $u_E = u_B$ The average energy density of an electromagnetic wave: $u = u_E + u_B$ $u = \frac{1}{2}\varepsilon_0 E_0^2 = \varepsilon_0 E_{rms}^2$ or $u = \frac{B_0^2}{2\mu_0} = \frac{B_{rms}^2}{\mu_0}$	B_{rms} = root mean square value of B. $E_{r.m.s}$ = root mean square value of E. u = average energy density of EM wave. u_E = energy density of electric field u_B = energy density of magnetic field

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8.	**Momentum carried by E.M wave**	Momentum: $p = \frac{U}{v}$	U = average energy of wave v = speed of EM wave
9.	**Intensity of E.M wave**	$\text{Intensity of a wave} = \frac{\frac{\text{Energy}}{\text{time}}}{\text{Area}} = \frac{\text{Power}}{\text{Area}}$ $I = u_{av}\, c = \varepsilon_0\, E_{rms}^2\, c$	I = Intensity of the wave u_{av} = average energy density of em wave E_{rms} = root mean square value of E